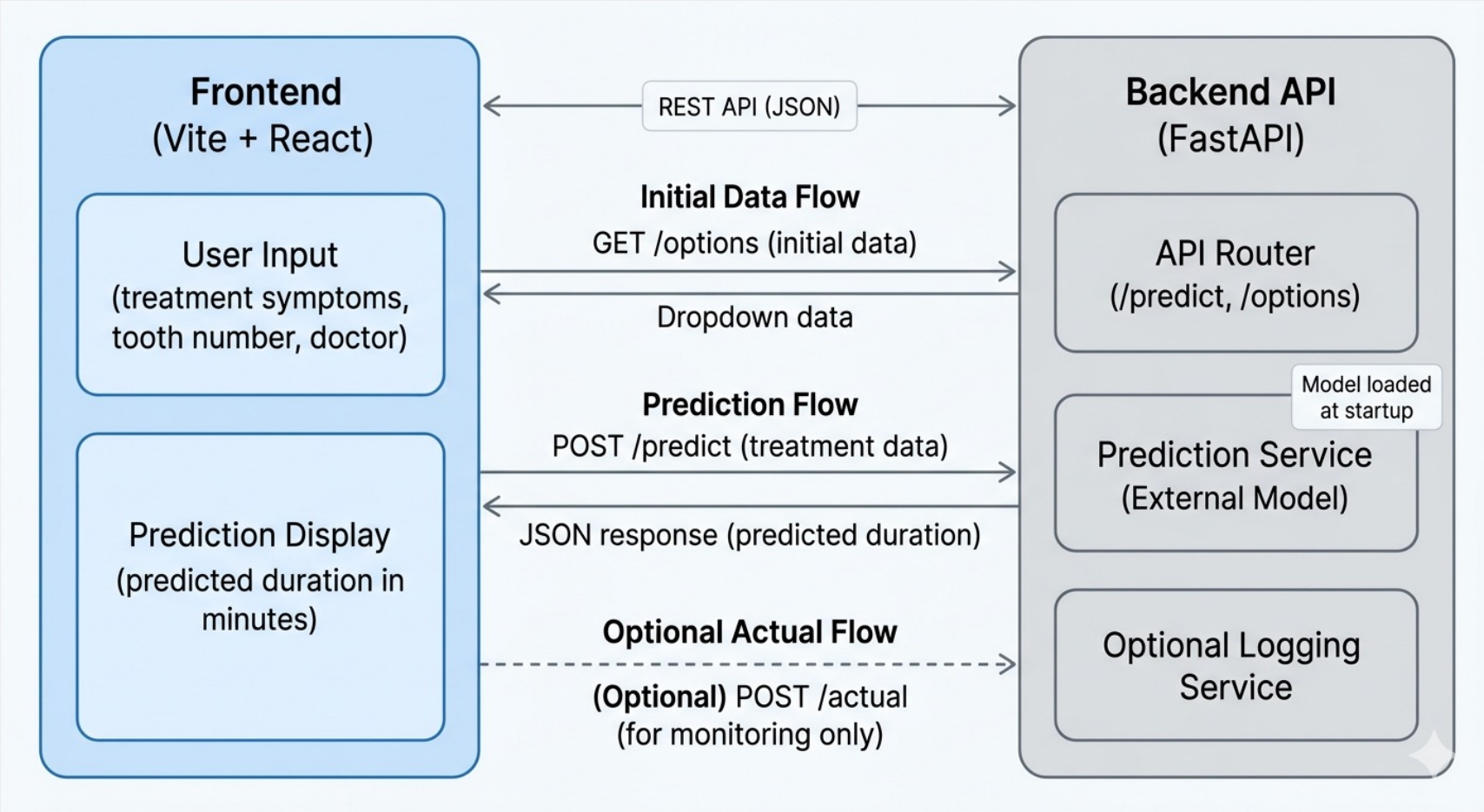
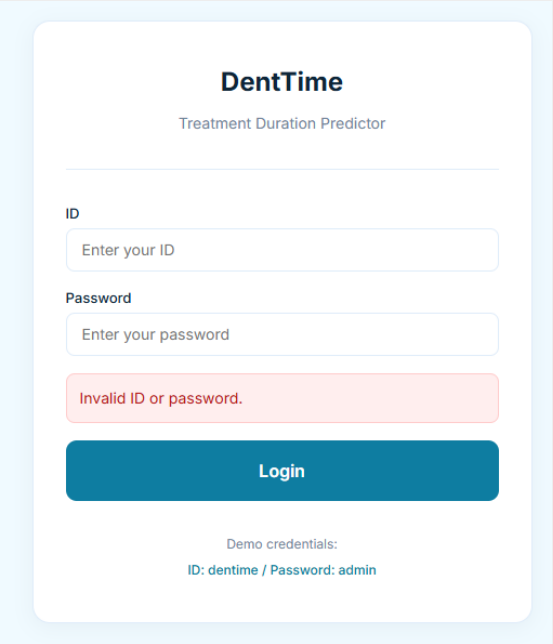
FRONTEND & BACKEND

SYSTEM PIPELINE

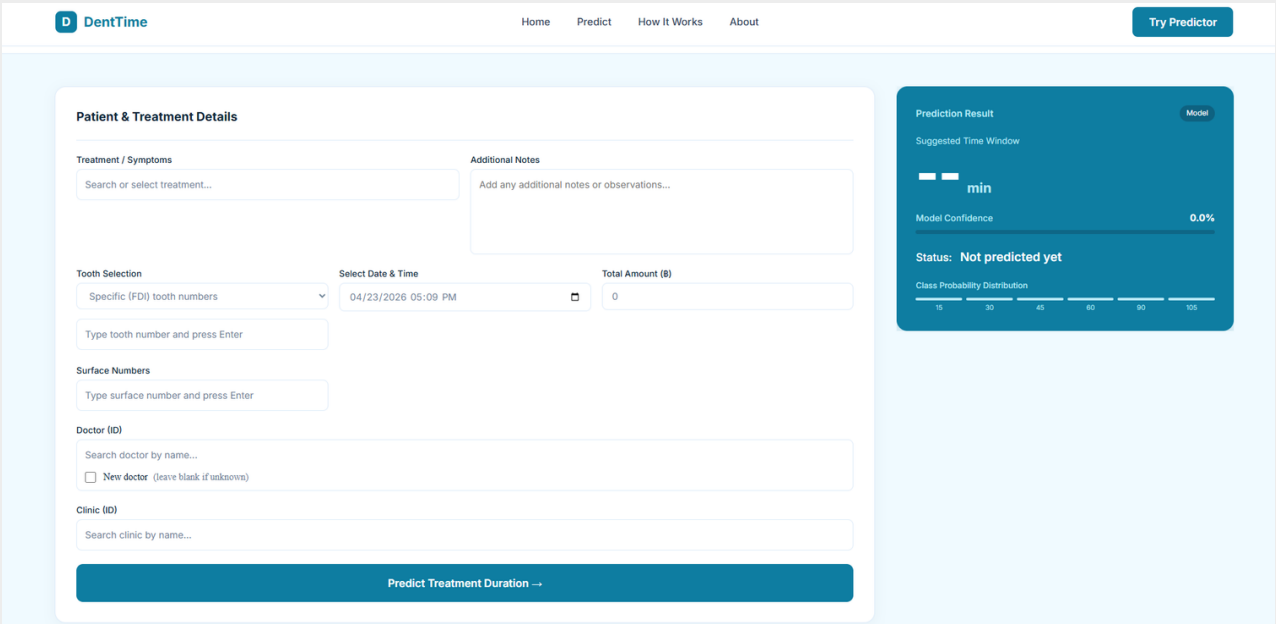


DENTIME

FRONTEND OVERVIEW



AUTHENTICATION PAGE



PREDICT PAGE

USER INPUT

Patient & Treatment Details

Treatment / Symptoms

Search or select treatment...

Additional Notes

Add any additional notes or observations...

Tooth Selection

Specific (FDI) tooth numbers

Type tooth number and press Enter

Select Date & Time

04/23/2026 05:09 PM

Total Amount (฿)

0

Surface Numbers

Type surface number and press Enter

Doctor (ID)

Search doctor by name...

☐ New doctor (leave blank if unknown)

Clinic (ID)

Search clinic by name...

Predict Treatment Duration →

POST /predict

DENTIME

MAPPING BEFORE GO IN MODEL

```python
class APIRequest(BaseModel):
    treatmentSymptoms: str
    toothNumbers: Optional[str] = None
    surfaces: Optional[str] = None
    totalAmount: float
    selectedDateTime: str
    doctorId: Optional[str] = None
    clinicId: str
    notes: Optional[str] = None
```

8 user input ⟶ 12 input

```python
class PredictRequest(BaseModel):
    clinic_pseudo_id: str
    dentist_pseudo_id: Optional[str] = None
    has_dentist_id: int                          # 0 or 1 not done
    treatment: str
    tooth_no: Optional[str] = None
    surfaces: Optional[str] = None
    total_amount: float
    has_notes: int                               # 0 or 1
    appt_day_of_week: int                        # 0-6
    appt_hour_bucket: int                        # 0 if unknown, else {4,8,12,16,20}
    is_first_case: int                           # 0 or 1
    appointment_rank_in_day: Optional[int] = None
```

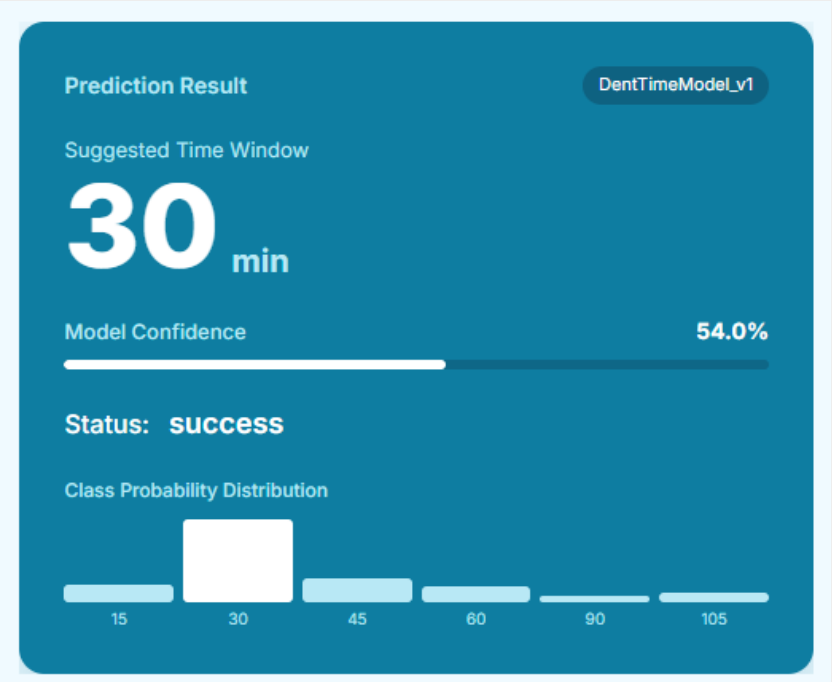
FEATURE TRANSFORMER

PREDICT

RETURN OUTPUT

DENTIME

PREDICT OUTPUT

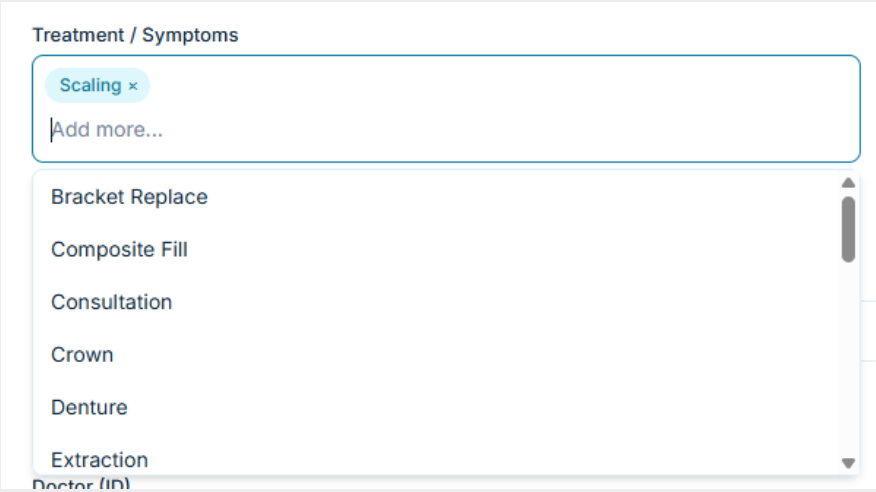


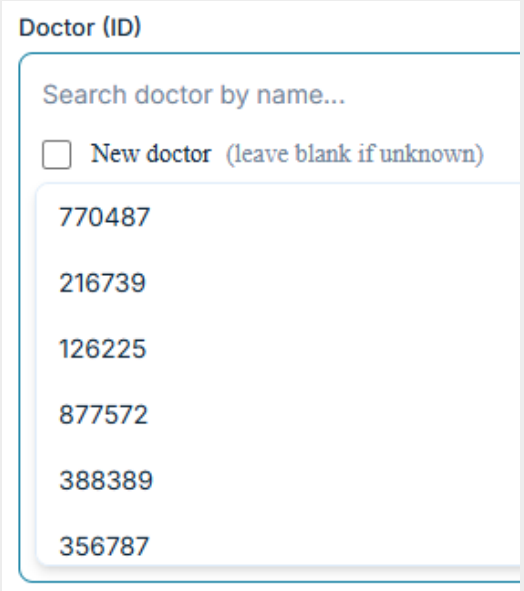
MAX(PROB) = 54.0 %

HOW DO WE GET DROPDOWN

GET /OPTION

Profile already load at start





EXAMPLE

Patient & Treatment Details

Treatment / Symptoms

Scaling × Consultation ×

Add more...

Additional Notes

โปร 888

Tooth Selection

Full mouth

Full mouth

Select Date & Time

04/23/2026 01:00 PM

Total Amount (฿)

1000

Doctor (ID)

Search doctor by name...

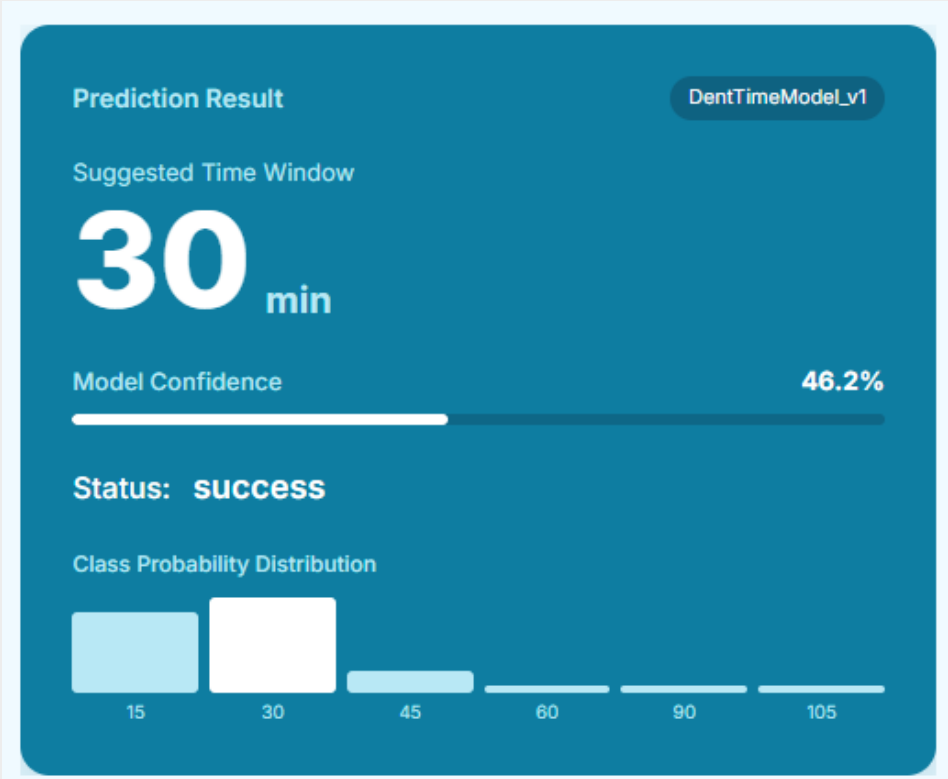
☐ New doctor (leave blank if unknown)

Clinic (ID)

538347

DENTIME

EXAMPLE



DENTIME

LIMITATION

- **Basic UI without advanced user experience design**
- **Backend uses simple API without full error handling coverage**
- **Limited scalability with current Docker-based setup**

FUTURE

- Improved UI/UX
- Optimized Backend Performance
- Prepare for scaling
- Collect /actual outcome